

Trustworthy AI in Avionics Systems: What Can We Learn from Certification Argumentation?

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Abstract—The increasing integration of machine learning into safety-critical avionics systems raises fundamental challenges for certification, particularly regarding trustworthiness, explainability, and evidence-based assurance. While initiatives from the European Union Aviation Safety Agency define anticipated Trustworthy AI objectives, these remain at a high level of description and lack operational guidance for requirements specification and certification.

This position paper investigates a research direction aimed at addressing this gap through the reuse of argumentation diagrams and pattern-based assurance techniques from safety-critical engineering. Building on established argumentation approaches, we explore how structured argumentation can support the decomposition of EASA Trustworthy AI objectives into traceable claims, warrants, and evidence.

As this work remains exploratory, we identify key research challenges, including pattern reproducibility across objectives, validation with certification stakeholders, tooling integration into RE and V&V workflows, and cross-domain transferability. We argue that argumentation-based Requirements Engineering techniques provide a promising foundation for bridging the gap between high-level Trustworthy AI principles and certifiable, evidence-backed requirements in avionics.

I. INTRODUCTION

The increasing adoption of *Artificial Intelligence* (AI) in safety-critical systems raises major challenges regarding trustworthiness, certification, and regulatory compliance. In avionics, these challenges are further amplified by the need to provide structured evidence supporting safety and assurance claims. As a result, *Trustworthy AI* has become a central concern for both certification authorities and industrial stakeholders.

Recent initiatives, including proposals from the *European Union Aviation Safety Agency* (EASA), define high-level objectives for Trustworthy AI in aeronautical systems [1]. However, these objectives remain generic, and their translation into operational, assessable, and evidence-based requirements remains largely open.

In this work, we focus specifically on the *technical* dimension of Trustworthy AI [2] in safety-critical avionics systems. Our objective is not to address broader ethical or societal concerns, such as fairness, value alignment, or perceived user trust, but to investigate how technical trustworthiness objectives, including robustness, traceability, behavioral consistency, and evidence-based assurance, can be operationalized within certification-oriented engineering processes. This distinction

is essential because certification activities rely fundamentally on explicit justification structures and verifiable evidence, as opposed to subjective notions of trust perception.

In this context, the *Requirements Engineering* (RE) community is well positioned to contribute, given its established methods for structuring goals, making assumptions explicit, and supporting traceability between requirements and evidence. More specifically, this paper positions Trustworthy AI as a Requirements Engineering problem of structuring assurance arguments. Safety-critical domains such as avionics already rely on argumentation-based approaches to organize certification evidence and support assurance processes. Approaches such as *Goal Structuring Notation* [3], [4], *Claim-Argument-Evidence* [5], and earlier work by Polacsek on *argumentation patterns* [6] illustrate how structured argumentation can support the construction of assurance cases.

We explore the hypothesis that such argumentation diagrams and reusable argument patterns could provide a generic structuring mechanism for organizing the body of evidence supporting Trustworthy AI objectives. In particular, they may help bridge the gap between high-level regulatory objectives and operational assurance artifacts.

Recent work in Requirements Engineering has begun to address the operationalization of Trustworthy AI concerns through structured, artifact-oriented approaches. Notably, Vilamizar et al. [7] propose integrating perspective-based concern elicitation with domain-independent RE artifacts to translate high-level Trustworthy AI principles into traceable requirements specifications. Their vision emphasizes the need for systematic mechanisms that bridge regulatory AI concerns, stakeholder viewpoints, and concrete engineering artifacts.

Our work addresses a complementary but distinct challenge. Rather than focusing on the *specification* of Trustworthy AI requirements themselves, we investigate how compliance with these objectives can be *justified* and *demonstrated* within safety-critical certification processes. Building on established assurance case methodologies [8], we explore how hierarchical argumentation structures and reusable justification patterns can organize claims, warrants, and domain-specific evidence to support EASA's high-level Trustworthy AI objectives in avionics. From an RE perspective, our contribution does not introduce a new requirements specification framework; instead, it proposes an argumentation-based structuring approach to make

assurance arguments for Trustworthy AI explicit, auditable, and systematically reusable across certification contexts.

Instead of proposing a complete solution, this paper outlines an initial research direction and ongoing exploratory work. We investigate whether argumentation patterns could support the operationalization and structuring of the high-level Trustworthy AI objectives proposed by EASA, and discuss how established assurance practices from avionics may inform future RE approaches for AI-based systems.

II. CONTEXT

The EASA has recently proposed a foundational framework addressing the integration of *Machine Learning* (ML) in safety-critical aeronautical systems. This initiative targets the limitations of traditional certification approaches, which were originally designed for deterministic and fully specified software, and are therefore not directly suited to data-driven systems whose behavior depends on training data, statistical generalization, and operational context.

To address these challenges, in its concept paper *Guidance for Level 1&2 Machine Learning Applications* [1], EASA identifies several key concerns, including model predictability, out-of-distribution behavior, explainability, cybersecurity of learning-enabled components, and the preservation of aviation-grade safety assurance. These concerns are organized under a Trustworthy AI paradigm, which aims at enabling the gradual integration of AI technologies while maintaining strong safety guarantees and human oversight.

A central outcome of this framework is the definition of high-level Trustworthy AI objectives structured across several domains, including safety assurance, data governance, and learning process management. While this represents an important step toward formalizing certification expectations for ML-based systems in aviation, these objectives remain mostly generic. As a consequence, their operationalization into verifiable requirements, as well as the systematic production of supporting evidence, is still an open challenge.

This gap raises two closely related needs: (i) the refinement of high-level Trustworthy AI objectives into operational and assessable requirements, and (ii) the explicit structuring of the corresponding body of evidence used to justify compliance. Addressing these needs requires suitable formalisms capable of linking objectives, requirements, assumptions, and evidence in a traceable and reviewable manner.

III. ARGUMENTATION DIAGRAMS AND PATTERN-BASED ASSURANCE

Argumentation diagrams [6] provide a structured visual framework to organize safety and certification arguments. They decompose a high-level *claim* (e.g., a certification objective or assurance goal) into a hierarchy of sub-claims, *strategies* (or *warrants*), and supporting *evidence*. In this model, a strategy explicates the reasoning or decomposition approach that links claims, while evidence refers to concrete artifacts (e.g., test reports, formal analyses, expert reviews) that substantiate them. This explicit separation ensures that

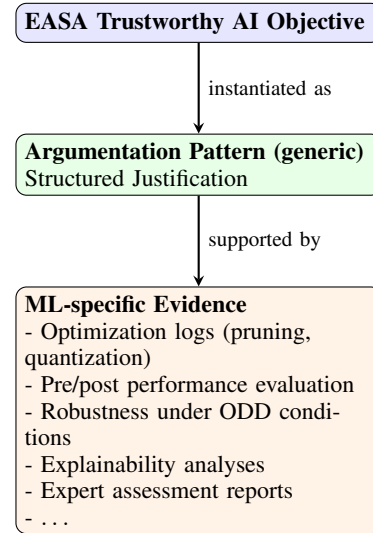


Fig. 1. Mapping EASA Trustworthy AI objectives to pattern-based argumentation and ML-specific evidence.

the rationale for trusting a system is transparent, auditable, and traceable.

Building assurance cases from scratch for each new standard or technology, however, is time-consuming, error-prone, and highly dependent on individual expertise. To address this, the certification community increasingly relies on *assurance case patterns*: generic, reusable templates that capture proven justification structures for recurring certification claims [9]. In this work, we adopt the pattern elicitation methodology proposed by Delmas et al. [8], which provides a systematic, domain-agnostic process for designing certification patterns. Indeed, they define an iterative and collaborative process for constructing such patterns with domain experts. This approach emphasizes the elicitation of explicit warrants, the avoidance of process-oriented reasoning biases, and the validation of patterns through expert review.

We hypothesize that such a pattern-based approach can be leveraged to structure Trustworthy AI objectives in avionics, as presented in Figure 1. In this perspective, high-level EASA objectives can be interpreted as top-level claims, that are then refined in sub-claims supported by evidences, in the form of AI-specific assurance artifacts, and progressively organized through reusable argument strategies.

IV. PRELIMINARY INVESTIGATION: APPLYING PATTERN ELICITATION TO EASA OBJECTIVES LM-01 AND LM-06

As an initial step toward evaluating our hypothesis, we applied the pattern elicitation methodology of Delmas et al. [8] to two EASA objectives, namely *LM-01* and *LM-06*. *LM-01* addresses the need to formally describe the ML model architecture prior to the training phase, yet leaves the expected granularity of the description intentionally open. *LM-06* focuses on post-training optimizations (e.g., pruning, quantization, embedded conversion), requiring rigorous documentation and impact assessment to ensure that such trans-

formations do not compromise model robustness, accuracy, or safety properties. Both objectives embody core Trustworthy AI challenges while remaining sufficiently abstract to warrant structured justification patterns.

Following Delmas et al.’s guidelines, we established two parallel design teams, each dedicated to one objective. Each team comprised a facilitator skilled in critical thinking and assurance case modeling, alongside domain experts in aeronautical AI. Over a series of iterative workshops, the teams progressively decomposed each high-level objective into hierarchical argumentation structures. Strict wording conventions were enforced to elicit explicit warrants and avoid process-oriented reasoning, while individual reflection phases between meetings helped surface missing evidence or hidden assumptions.

The investigation remains explicitly preliminary. Both teams are currently in the active pattern drafting phase, and it is too early to assess definitive stability or success¹. Nevertheless, early results are highly encouraging: the structured decomposition has already clarified implicit certification expectations and identified concrete evidence. Our immediate next steps are to finalize the draft patterns, subject them to validation by experts, and evaluate their operational usefulness in real-world.

V. DISCUSSION AND RESEARCH CHALLENGES

This preliminary investigation raises three research questions that we plan to address in future work.

Validation through cross-disciplinary collaboration. Pattern construction is sensitive to expert bias and incomplete domain coverage. Robust validation therefore requires structured engagement with certification authorities, AI practitioners, and safety engineers. Iterative elicitation, peer review, and scenario-based evaluation are necessary to uncover missing warrants and strengthen argument completeness.

Tooling and integration. Practical adoption hinges on seamless integration with existing requirements management and verification workflows. As demonstrated by artifact-centric approaches like AMDiRE [7], tool-supported operationalization is critical to guide stakeholders, navigate trustworthiness concerns, and maintain end-to-end traceability across development artifacts. Similarly, Delmas et al. [8] emphasize that assurance pattern elicitation and instantiation are complex, iterative processes that require structured environments to manage warrants, track evidence, and support cross-disciplinary validation. Future work should explore how pattern repositories can be linked to tools that automate traceability, support impact analysis, and ease certification audits. A first step in this direction is the `jpipe-compiler` project², which aims to provide a compiler-based infrastructure for managing justification patterns and their instances. Investigating how such tooling can support *living assurance cases* for evolving

AI models, while interoperating with existing RE and ML engineering pipelines, is a promising research avenue.

Cross-domain transferability. While this work focuses on avionics, similar Trustworthy AI concerns arise in domains such as automotive systems (ISO 21448 :2022 Safety Of The Intended Functionality) [10] or medical devices (AIB 2025-1) [11]. A key question is whether elicited argumentation patterns are transferable across domains or require significant adaptation. Addressing this question could support the development of a reusable library of Trustworthy AI assurance patterns. In addition, recent discussions in Trustworthy AI literature highlight the limitations of checklist-based compliance and advocate for contextualized, systemic assurance mechanisms [2]. From this perspective, reusable argumentation patterns offer a structured alternative to superficial verification by explicitly linking high-level objectives to domain-specific evidence through transparent warrants and assumptions. Investigating how these patterns adapt across safety-critical domains thus emerges as a key research direction for scalable and auditable AI assurance.

Overall, addressing these challenges is essential to move from exploratory argumentation structures toward reusable and operational RE artifacts for Trustworthy AI in safety-critical systems.

VI. CONCLUSION

This position paper investigates a research direction at the intersection of Requirements Engineering and Trustworthy AI certification in avionics. It explores the reuse of argumentation diagrams and pattern-based assurance techniques to support the operationalization of EASA high-level Trustworthy AI objectives. We initiated a preliminary exploration centered on two EASA objectives, one addressing the description of ML Model Architecture, and the other addressing post-training optimizations and their impacts.

The contribution of this work is not a finalized framework, but a Requirements Engineering perspective that reinterprets Trustworthy AI objectives as structured, pattern-driven assurance cases. In this view, argumentation patterns provide a potential mechanism to bridge the gap between high-level regulatory objectives and traceable, evidence-based requirements in safety-critical systems. We position this work as an invitation to further explore pattern-based argumentation as a RE mechanism for structuring Trustworthy AI arguments and supporting the development of certifiable AI-enabled systems.

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¹A first draft is available at https://thomaspolacsek.net/LM-01_ArgumentationPattern.pdf. We plan to upload a final version of the LM-01 pattern to arXiv before the workshop and, perhaps, the pattern for the LM-06 as well.

²www.jpipe.org

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